

- Q.4 Which of the following is not a type of actuator? (CO4)
- a) Electric motor b) Pneumatic cylinder
- c) Hydraulic motor d) Gear pump
- Q.5 Fluid power circuits use symbols standardized by (CO4)
- a) ISO b) IEEE
- c) ANSI d) BIS
- Q.6 A flow control valve affects (CO3)
- a) Pressure b) Direction
- c) Speed d) Volume

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define hydraulic power. (CO1)
- Q.8 What is the main advantage of pneumatics over hydraulics? (CO5)
- Q.9 Name two types of actuators used in fluid power systems. (CO4)
- Q.10 List any two control elements in pneumatic systems. (CO3)
- Q.11 What is meant by fluid power? (CO1)
- Q.12 Name one safety device used in hydraulic circuits. (CO2)

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SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the differences between hydraulic and pneumatic systems. (CO5)
- Q.14 Describe the working of a pressure relief valve. (CO2)
- Q.15 Explain with a diagram the construction of a double-acting cylinder. (CO5)
- Q.16 Describe the function of a flow control valve. (CO3)
- Q.17 Draw and explain a basic hydraulic circuit. (CO5)
- Q.18 Describe the working of a pneumatic circuit used for clamping. (CO2)
- Q.19 Explain any four ISO symbols used in fluid circuits. (CO2)
- Q.20 Describe the maintenance needs of fluid power systems. (CO4)
- Q.21 Compare hydraulic motors with pneumatic motors. (CO3)
- Q.22 Describe any one application of a fluid power system in automation. (CO2)

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